Utility Theory and Decision Networks

Lecture 22

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Last Time: The Robot Estimated Its State

Dynamic Bayesian networks and filters gave the robot a belief about the current world:

$$bel_t(x) = P(X_t = x \mid z_{1:t}).$$

- HMMs represented compact discrete hidden state.
- Kalman filters represented linear-Gaussian continuous state.
- Particle filters represented flexible beliefs using weighted samples.

Symbols: $bel_t(x)$: belief assigned to candidate state x at time t; X_t: hidden state; $z_{1:t}$: observations through t; $P(\cdot \mid \cdot)$: probability given evidence. Known control history is suppressed.

Key idea

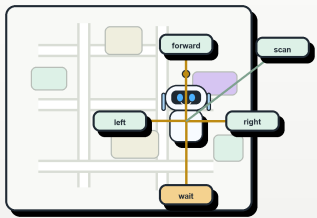
State estimation gives the agent beliefs about the world; decision theory uses those beliefs to choose actions.

The Robot Must Choose Among Route Options

A planner proposes options; the robot must choose which option to execute under uncertainty.

The choice depends on:

- current position belief;
- obstacles and pedestrian traffic;
- remaining time and battery;
- delivery priority.



Key idea

Estimate the situation, choose an action, observe the result, and repeat.

A Good State Estimate Still Does Not Tell Us What to Do

Let B describe whether the shortcut is blocked:

$$P(B = 1) = 0.35, \qquad P(B = 0) = 0.65.$$

Throughout the example, write these as $P(\text{Blocked})$ and $P(\text{Clear})$. That belief does not answer:

- Is the shortcut worth attempting?
- Should the robot take a longer detour?
- Is another scan worth its time and energy?

Symbols: B: blockage variable; 1: blocked, 0: clear. P: probability given the current evidence, which is suppressed here.

Key idea

Probability describes what may happen. Choosing an action also requires preferences over the consequences.

The New Control Loop



The filter supplies a belief; the decision model uses it to compare actions.

Symbols: b_t: current belief distribution at time t; it summarizes the observations and known controls so far.

Key idea

Today we solve one decision at a time. Later we will account for how an action changes future choices.

At One Campus Intersection, the Robot Has Several Actions

Shortcut

Fastest route. May be blocked by a crowd or maintenance cart.

Detour

Longer route around the building. Usually clear and predictable.

Wait

Spend time while congestion may clear.

Scan

Pause for a better sensor reading before committing.

Key idea

An action is valuable because of the distribution of outcomes it creates—not because its name sounds desirable.

First, Isolate One Route Decision

We will compare **shortcut** and **detour** at the same intersection.

- Blockage is hidden; choosing a route does not change whether the shortcut is currently blocked.
- Utilities include the time, energy, and risk of completing this route segment.
- The detour has a fixed utility in our simplified model.
- Later, we will allow one scan before choosing the route.

Waiting changes the world over time; we leave that extension for sequential decision models.

Key idea

Keep the decision, uncertainty, and outcome boundary clear before calculating anything.

Utility Turns Preferences into a Numerical Model

A **utility function** assigns a value to each outcome:

$$U : \mathcal{O} \to \mathbb{R}.$$

For example,

$$U(\text{fast safe delivery}) > U(\text{safe detour}) > U(\text{blocked attempt}).$$

For risky choices, the numerical gaps must also encode willingness to trade a chance of a good outcome against a chance of a bad one.

Symbols: $\mathcal{O}$: set of outcomes; o: one outcome; $U(o)$: its utility; $\mathbb{R}$: real numbers; $>$: preferred to.

Key idea

Ranking certain outcomes is only the start. Expected utility needs a scale that also represents preferences over uncertain outcomes.

Under Uncertainty, Use Expected Utility

Average the utility of each possible outcome, weighted by its probability:

$$EU(a \mid e) = \sum_{o} P(o \mid a, e) U(o).$$

Then choose among feasible actions:

$$a^* = \arg\max_{a \in \mathcal{A}} EU(a \mid e).$$

Symbols: a: candidate action; e: evidence already observed; o: possible outcome; $P(o \mid a, e)$: outcome probability under the chosen action and evidence. $\sum_o$: sum over outcomes; $\mathcal{A}$: feasible action set; $\arg\max$: return a maximizing action; a^*: chosen action.

Key idea

Probability supplies the weights; utility supplies the values. We will suppress fixed evidence e in the numerical example.

Worked Decision: Shortcut or Detour?

Current belief: $P(\text{Clear}) = 0.65$, $P(\text{Blocked}) = 0.35$.

Utility $U(a, b)$	**Clear** ($b = 0$)	**Blocked** ($b = 1$)
Shortcut	80	-60
Detour	45	45

These are illustrative preference values, not measured travel times. The detour is unaffected by shortcut blockage in this model.

Symbols: a: route action; b: value of blockage variable B; $U(a, b)$: utility of the resulting route-segment outcome. Higher is preferred; negative values indicate outcomes worse than the chosen zero.

Key idea

Compare both actions under the same belief and the same utility scale.

Expected Utility of the Shortcut

$$EU(\text{Shortcut}) = 0.65(80) + 0.35(-60)$$
$$= 52 - 21 = 31.$$

The shortcut is more likely to be clear than blocked, but the blocked outcome is costly enough to reduce its expected utility substantially.

Symbols: $EU(\text{Shortcut})$: expected utility of the shortcut; $0.65, 0.35$: clear/blocked probabilities; $80, -60$: their utilities. Each product is probability $\times$ utility; 31: their sum.

Key idea

Expected utility combines the upside and downside of an action in one probability-weighted quantity.

Expected Utility of the Detour

For the simplified example, the detour is predictable:

$$EU(\text{Detour}) = 45.$$

Therefore

$$45 > 31.$$

The rational one-step choice is the detour.

Symbols: $EU(\text{Detour})$: expected utility of the detour, 45 in both world states. 31: shortcut value; $45 > 31$: detour has greater expected utility under the same belief.

Key idea

Rational choice is about expected preference, not simply choosing the most probable outcome.

When Would We Switch to the Shortcut?

Let p be the probability that the shortcut is clear. It beats the detour when

$$\underbrace{80p - 60(1-p)}_{EU(\text{Shortcut})} > \underbrace{45}_{EU(\text{Detour})}.$$

Rearranging gives

$$140p > 105 \qquad \Longrightarrow \qquad p > 0.75.$$

At $p = 0.75$ the actions tie; at our current $p = 0.65$, choose the detour.

Symbols: $p = P(\text{Clear})$; $1 - p = P(\text{Blocked})$; $80, -60, 45$: utilities from the table. EU: expected utility; $>$: strict preference here; $\Longrightarrow$: implies.

Key idea

"More likely clear than blocked" is insufficient. These utilities require more than a 75% chance of a clear shortcut.

Changing the Utility Scale Preserves the Decision

Under the standard expected-utility consistency assumptions on preferences over lotteries, utilities represent choices under uncertainty.

$$U'(o) = \alpha U(o) + \beta, \qquad \alpha > 0.$$

Then

$$EU'(a) = \alpha EU(a) + \beta.$$

A positive scaling and a common shift preserve which action has the greatest expected utility.

Symbols: o: outcome; a: action; U, U': original/transformed utilities; EU, EU': corresponding expected utilities. α: positive scale; β: common offset. Outcome probabilities sum to 1. A lottery is a probability distribution over outcomes.

Key idea

Arbitrary increasing transformations preserve rankings of certain outcomes, but can change decisions under uncertainty. Utility gaps matter.

A Robot Usually Has More Than One Objective

The robot may simultaneously care about:

- travel time;
- energy consumption;
- collision and pedestrian risk;
- progress toward completing the delivery;
- reliability and schedule slack.

These objectives can conflict.

Key idea

Real decision systems often optimize several competing attributes, so the utility model must make those tradeoffs explicit.

One Engineering Approximation: Multiattribute Utility

Combine attribute scores on stated, comparable scales:

$$U(o) = w_t u_t(o) + w_e u_e(o) + w_s u_s(o) + w_p u_p(o).$$

Symbols: o: outcome; u_t, u_e, u_s, u_p: scores for time, energy, safety, and progress, all oriented so higher is better. w_t, w_e, w_s, w_p: nonnegative tradeoff weights; here subscripts identify attributes, not time steps.

A larger weight matters relative to the score's scale: changing a score's units requires adjusting its weight.

Key idea

An additive model assumes tradeoffs can be separated this way. Attribute interactions may require a richer utility model.

Changing the Weights Can Change the Robot's Behavior

Time-heavy

Large penalty for delay. The robot favors feasible routes with smaller expected delay.

Energy-heavy

Battery is scarce. The robot avoids steep ramps, long scans, and unnecessary detours.

Safety-heavy

The robot prefers conservative routes through uncertain pedestrian traffic.

Key idea

The same probability model can produce very different behavior under different utility functions.

Not Every Requirement Should Be “Just Another Weight”

Weighted utilities make tradeoffs possible by design.
But some requirements may instead be treated as **hard constraints**:

- exclude actions that violate safety envelopes;
- enforce legal or operational limits;
- optimize utility only among feasible actions.

This matters when a tiny probability of a catastrophic outcome should not be casually exchanged for a small improvement in speed.

Key idea

Utility maximization is only as sensible as the objectives, constraints, and action set supplied to it.

Decision Networks: Three Node Types

A **decision network** or **influence diagram** contains three node types:

Chance node

Uncertain random variable.
Example: shortcut blocked?

Decision node

Choice controlled by the agent.
Example: shortcut, detour, scan, wait.

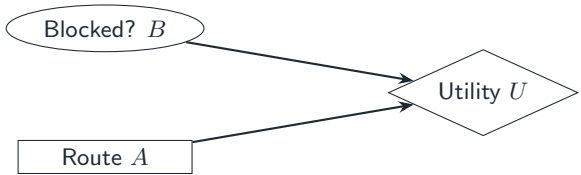
Utility node

Preference over resulting outcomes.
Example: time + energy + safety + progress.

Key idea

A decision network combines uncertainty, controllable choices, and utility in one graphical model.

Robot Decision Network: One Route Choice



The utility table depends on the chosen route and the hidden blockage state. There is no arrow from B into A: blockage is not directly observed.

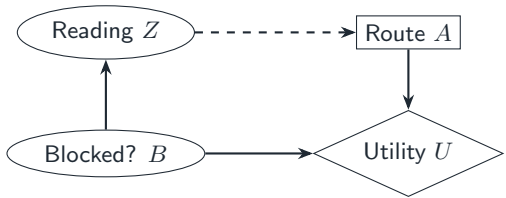
Symbols: B: chance variable (ellipse); A: controlled choice (rectangle); $U(a, b)$: utility function (diamond), not a probability table. Arrows into U identify its inputs.

Key idea

Choose a route using a belief about blockage, without pretending the hidden state is known.

Information Arcs Mean "Known Before the Decision"

Suppose a scan reading is available before the route is chosen.



Symbols: B: hidden blockage; Z: noisy sensor reading; A: route choice; U: utility. Solid arrows into chance nodes define probability models; the dashed arrow into A indicates available information.

Key idea

A decision node receives information, not a CPT. The reading is known; the blockage state remains uncertain.

Evaluate Using Only Information Available at Decision Time

After a reading z, average over the still-hidden blockage state:

$$EU(a \mid z) = \sum_{b \in \{0,1\}} P(B = b \mid Z = z) U(a, b).$$

Then select a route for that reading:

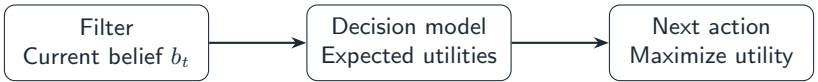
$$a^*(z) = \arg \max_{a \in \{\text{Shortcut}, \text{Detour}\}} EU(a \mid z).$$

Symbols: b: possible blockage value; z: observed reading; $P(B = b \mid Z = z)$: posterior belief. $U(a, b)$: utility table; $a^*(z)$: decision rule mapping a reading to a route. Fixed earlier evidence is suppressed.

Key idea

Average over hidden states before maximizing. Maximizing separately for each hidden state would incorrectly give the robot perfect information.

The Belief State Plugs Directly Into the Decision



The filter updates what the robot believes. The decision model uses that distribution to compare consequences of feasible actions.

Symbols: b_t: current belief distribution at time t. In the worked example it supplies the probabilities of clear and blocked.

Key idea

State estimation and action selection are complementary steps in the same controller.

One-Step Decision Procedure

At each control cycle:

1. update the belief state b_t;
2. generate the feasible action set $\mathcal{A}(b_t)$;
3. predict possible outcomes for each action;
4. compute expected utility for each action;
5. execute a maximizing feasible action.

Symbols: b_t: current belief at time t; $\mathcal{A}(b_t)$: actions allowed under that belief and the operational constraints; expected utility averages their outcome utilities.

Key idea

For movement, this compares one segment at a time. A scan needs a short lookahead: scan, observe, then choose a route.

Value the Plan: Scan, Observe, Then Choose

Compare the best route now with the best route after each possible reading:

$$V_{\text{now}} = \max_a EU(a \mid e), \qquad V_{\text{with}} = \sum_z P(z \mid e) \max_a EU(a \mid e, z).$$

The **gross value of information** is $V_{\text{with}} - V_{\text{now}}$. Acquire the reading when this improvement exceeds its acquisition cost.

Symbols: e: current evidence; a: route choice; z: possible reading; EU: expected utility. V_{now}: optimal value before sensing; V_{with}: expected optimal value after sensing, before its cost. Sum over readings; choose the route after each reading.

Key idea

A scan's immediate cost buys a later decision. Evaluating only its immediate reward would miss its benefit.

Will a Better Measurement Change the Decision?

The prior probability of blockage is $P(\text{Blocked}) = 0.35$. Without scanning, choose the detour: $V_{\text{now}} = 45$.
A scan costs $c = 5$ utility units and has the sensor model

$$P(+ \mid \text{Blocked}) = 0.90, \qquad P(+ \mid \text{Clear}) = 0.10.$$

Symbols: $+$: sensor reports blockage; $-$: sensor reports clear. These are readings, not true states. c: fixed additive utility cost; V_{now}: value of the best route without another reading.

Assume the world stays fixed during this brief scan, both routes remain available, and its delay and energy cost are captured by c.

Key idea

We are valuing one imperfect measurement before one route choice, using the same utilities as before.

The Scan Produces Two Possible Belief Updates

Weight each sensor likelihood by its prior state probability:

$$P(+) = 0.90(0.35) + 0.10(0.65) = 0.38, \qquad P(-) = 1 - 0.38 = 0.62.$$

Bayes' rule gives

$$P(\text{Blocked} \mid +) = \frac{0.90(0.35)}{0.38} \approx 0.829,$$

$$P(\text{Blocked} \mid -) = \frac{0.10(0.35)}{0.62} \approx 0.0565.$$

Symbols: $P(+), P(-)$: reading probabilities; $P(\text{Blocked} \mid +)$ and $P(\text{Blocked} \mid -)$: posterior blockage beliefs. The negative-reading likelihood under blockage is $1 - 0.90 = 0.10$; denominators normalize each posterior.

Key idea

Predict how often each reading occurs, then update the belief separately for each branch.

After the Scan, Choose a Route for Each Reading

For a **positive** reading, blockage probability is about 0.829:

$$EU(\text{Shortcut} \mid +) \approx (1 - 0.829)(80) + 0.829(-60) \approx -36.1.$$

Choose the **detour**, whose utility is 45.

For a **negative** reading, blockage probability is about 0.0565:

$$EU(\text{Shortcut} \mid -) \approx (1 - 0.0565)(80) + 0.0565(-60) \approx 72.1.$$

Choose the **shortcut**, since $72.1 > 45$.

Symbols: $EU(a \mid +)$ and $EU(a \mid -)$: route values after the reading, before sensing cost. $80, -60$: shortcut utilities; 45: detour utility. Displayed posteriors are rounded; calculations use full precision.

Key idea

The conditional plan is: positive $\rightarrow$ detour; negative $\rightarrow$ shortcut.

Separate Plan Value, Information Value, and Cost

Average the value of the best route in each reading branch:

$$V_{\text{with}} \approx \underbrace{0.38(45)}_{+:\ \text{detour}} + \underbrace{0.62(72.1)}_{-:\ \text{shortcut}} \approx 61.8.$$

Gross information value: $61.8 - 45 = 16.8$.
Scan-plan value after cost: $61.8 - 5 = 56.8$.
Net improvement over acting now: $56.8 - 45 = 11.8$.

Symbols: V_{with}: expected route utility after sensing, before cost. $0.38, 0.62$: reading probabilities; $45, 72.1$: optimized branch values; 5: scan cost. All values use the same utility scale.

Key idea

Choose the scan plan: its expected value 56.8 exceeds the best immediate route value 45 by 11.8.

Information Can Be Accurate and Still Have Zero Decision Value

Observation changes the action

Positive → detour
Negative → shortcut
The information can have positive decision value.

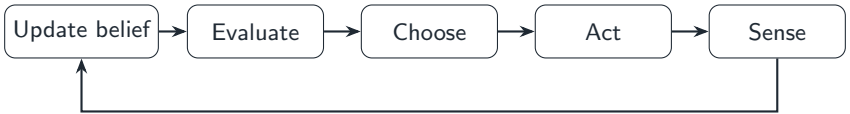
Same action either way

Positive → detour
Negative → detour
If the same action is optimal for every possible reading, gross information value is exactly zero.

Key idea

Optional information has nonnegative gross value because it can be ignored. Acquisition cost can make its net value negative.

Now Put the Decision Back Inside the Robot Loop



After acting, the robot receives new evidence and reevaluates its options. Repeating this loop does not by itself add long-term planning.

Key idea

The decision model can be reevaluated each cycle. Its planning horizon determines whether it is myopic.

This Gives Us a Myopic Decision Controller

One-step controller

```
while delivery not complete:
  b_t = update_belief(sensor_data)
  actions = feasible_actions(b_t)
  for a in actions:
    EU[a] = expected_immediate_utility(a,b_t)
  execute(argmax_a EU[a])
```

Here `b_t` is the current belief, `a` a feasible action, and `EU[a]` its expected immediate utility.
This version is **myopic**: it omits consequences beyond the immediate segment. To include sensing, evaluate the scan–then–route plan from the previous example, rather than just the scan cost.

Key idea

Repeated one-step optimization is a useful reactive strategy, but repeated local optimality does not guarantee globally good behavior.

Failure Mode: Locally Rational Actions Can Oscillate

Imagine two similar corridors whose estimated congestion changes slightly with each noisy scan.

1. At time t, left has slightly higher immediate expected utility $\rightarrow$ turn left.
2. At time $t+1$, a new reading makes right slightly better $\rightarrow$ turn right.
3. At time $t+2$, noise flips the preference again $\rightarrow$ turn left.

$$\text{Left} \leftrightarrow \text{Right} \leftrightarrow \text{Left} \leftrightarrow \cdots$$

Symbols: $t, t+1, t+2$: successive control cycles; $\rightarrow$: controller chooses that direction; $\leftrightarrow$: repeated switching, not a probabilistic dependency.

Key idea

A controller that only asks "what is best this instant?" can thrash when actions have switching costs or delayed effects.

Failure Mode: The Best Immediate Move Can Create a Bad Future

Narrow shortcut

Immediate utility: $+30$ (new example).
Moves quickly toward the destination, but enters a courtyard with one exit and frequent congestion.

Wider detour

Immediate utility: $+20$ (new example).
Looks worse right now, but preserves several future route choices.

These illustrative values concern only the immediate move; later congestion is omitted.

Key idea

Actions do not merely produce immediate outcomes—they change which states and choices will be available later.

The Question Has Changed Again

So far the robot chooses

$$a_t^* = \arg \max_{a \in \mathcal{A}(b_t)} EU(a \mid b_t).$$

But repeated autonomous control asks something stronger:

What action is best when I also care about the future states and future decisions it creates?

Symbols: t: current time; b_t: current belief; $\mathcal{A}(b_t)$: feasible actions; $EU(a \mid b_t)$: utility expected under that belief for the current outcome horizon; a_t^*: maximizing action.

Key idea

Once actions influence future opportunities, rational decision making must reason across sequences of states and actions.

Next: Markov Decision Processes

Lecture 23 will first assume the state is fully observed and formalize:

- **states**: what situation is the robot in?
- **actions**: what can the robot do?
- **transitions**: where might each action take it?
- **rewards**: what immediate signal does a transition produce?
- **policy**: what should the robot do in every state?

Key idea

Our one-decision examples become sequential planning problems. MDPs use observed states; partially observable MDPs (POMDPs) use hidden states and beliefs. Decision networks can also represent multiple decisions.

The Whole Unit Now Forms One Pipeline



Beliefs describe uncertainty. Utility describes preferences. A decision model joins them to select actions; sequential models account for what comes afterward.

Key idea

The course adds beliefs, time, preferences, actions, and finally long-term strategy.

Takeaways

- Probability describes uncertain outcomes; utility describes preferences over those outcomes.
- Expected utility combines probability and preference: choose the action with maximum expected utility.
- Multiattribute utilities make tradeoffs explicit, but weights and modeling assumptions directly shape behavior.
- Decision networks combine chance variables, decision variables, information, and utility.
- Value of information asks whether better evidence is worth its sensing cost because it changes downstream decisions.
- Repeating immediate-utility choices gives a myopic controller; modeling future choices requires an explicit planning horizon.

Key idea

The robot can now decide what to do next. Lecture 23 asks how it should decide when "next" changes everything that comes after.